

Inside CryoCell

The biophysics and governing equations behind a virtual mesenchymal stem cell for in-silico cryoprotectant testing.

Version 0.6 Calibration hMSC + DMSO, 1 °C/min → 35.4% S₂₄

Engine multi-compartment ODE, explicit integration Scope slow-freeze cryopreservation

ABSTRACT

CryoCell simulates a single human mesenchymal stem cell through a complete cryopreservation protocol — CPA loading, cooling, ice seeding, storage, warming, dilution and recovery — from first-principles transport and thermodynamics rather than fitted survival curves. Water and cryoprotectant cross the plasma membrane by Kedem–Katchalsky flux on a deforming, area-conserving membrane; the extracellular solution freeze-concentrates along a non-ideal liquidus; intracellular ice nucleates by surface-catalysed nucleation when the cytosol supercools faster than it can dehydrate; and injury accumulates through five biophysical channels plus a delayed apoptotic wave. Every visual element in the interface is a projection of a state variable the engine actually integrates. This paper documents the governing equations, their literature basis, and how each maps to what is drawn on screen.

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01 Architecture & state

CryoCell integrates one cell as a set of coupled compartments — cytosol, plasma membrane, nucleus, mitochondria, and the extracellular solution with its ice field — stepped through a nine-phase protocol.

The engine advances an explicit time integrator with flux caps (half/double-step drift < 0.3 survival points at the operating points). At each step it records a state vector; the interface reads that vector and draws it. The principal state variables are:

SYMBOL	STATE	MEANING
V, V_n	cell volume	absolute and normalized to isotonic (V/V_{iso})
C_{in}, C_{out}	CPA molarity	intracellular and extracellular cryoprotectant
O_{sme}	osmolality	extracellular total solute (osmol/L); freeze-concentrates
$f_{Ice}, grain$	ice field	extracellular ice fraction and mean grain size (μm)
dT_{sc}	supercooling	intracellular liquidus minus temperature ($^{\circ}C$)
P_{iif}	IIF probability	cumulative intracellular-ice nucleation probability
$gel, glass$	phase state	membrane gel fraction; cytosol vitrified fraction
$mt, actin, intf$	cytoskeleton	polymerized fraction of MT / F-actin / vimentin
$d\psi, mpt$	mitochondria	membrane potential $\Delta\psi_m$; permeability-transition
$pore, prot$	integrity	membrane defect fraction; native-protein fraction

Survival is reported at three horizons — S_{imm} (immediate), S_{24} (24 h) and S_{72} (settled, after the delayed death wave) — plus a functional-recovery index F_{rec} .

02 Membrane transport

Water and permeant CPA cross the membrane by the three-parameter **Kedem–Katchalsky** formalism, which — unlike a two-parameter (L_p, P_s) model with implicit $\sigma = 1$ — times the CPA volume excursion correctly.^{1,2} Volume flux is driven by the non-permeant osmotic gradient plus the reflected permeant gradient:

$$J_v = L_p \cdot A \cdot R \cdot T \cdot [(\Delta C_{np}) + \sigma \cdot (\Delta C_p)]$$
(1)

The permeant (CPA) flux carries a diffusive Fick term *and* a solvent-drag term proportional to the water flux:

$$J_s = P_s \cdot A \cdot (C_{out} - C_{in}) + (1 - \sigma) \cdot \bar{C} \cdot J_v \quad (2)$$

Membrane area is **not** constant: a dehydrated cell has less surface, so flux falls with it, while a folded-membrane reserve buffers the first ~40% of shrinkage before true area is lost:

$$A(t) = A_0 \cdot \max(V/V_{iso}, 0.60)^{2/3} \quad (3)$$

Each CPA carries its own reflection coefficient σ and permeation activation energy $E_{a,Ps}$ (Arrhenius-scaled with temperature), so the model ranks glycerol, DMSO, ethylene glycol and propylene glycol by their true relative permeability rather than a single generic constant. In the interface, equation (1) is what the **water-flux arrows** draw — outward as the cell dehydrates, inward on dilution — and the resulting V/V_{iso} is the **volume trace** on the timeline.

PROVENANCE • RE-FIT

hMSC L_p is set to 0.22 $\mu\text{m}/\text{min}/\text{atm}$ (within the measured 0.2–0.4 range) so the corrected variable-area model reproduces the inverted-U cooling-rate optimum. The earlier 0.15 had the constant-area error folded into it.

03 Solution thermodynamics

Freezing-point depression uses the **osmotic virial equation** of Elliott et al.,^{3,4} non-ideal and composition-specific, rather than the ideal-dilute limit:

$$\Delta T_f = K_f \cdot \text{osm} \cdot (1 + B \cdot \text{osm}) , \quad K_f = 1.86 \text{ } ^\circ\text{C} \cdot \text{kg}/\text{mol} \quad (4)$$

The second virial coefficient B is blended from the salt and CPA coefficients by osmolal weight. Below the freezing point, the extracellular solution sits *on the liquidus*: as pure water leaves as ice, the residual unfrozen fraction concentrates to stay in equilibrium. The **freeze-concentration factor** follows directly:

$$cf = \text{osm}_{\text{tot}}(T) / \text{osm}_0 \quad \Rightarrow \quad f_{\text{ice}} = 1 - 1/cf \quad (5)$$

so every extracellular species — salt, CPA, additive — scales by **cf**, and the ice volume fraction is fixed by how far the temperature has pushed along the liquidus. This is the **solution-effects** arm of Mazur's two-factor hypothesis: at slow cooling the cell is exposed for a long time to an increasingly concentrated brine.⁵ In the interface, equation (5) drives the density of the **extracellular CPA/salt field** in the unfrozen channel and the operating point on the **phase-diagram inset**.

Composition-specific glass-transition (T_g') and eutectic temperatures per CPA feed the mobility term (§5), so glycerol arrests far warmer than DMSO.⁶

04 Ice & nucleation

The fast-cooling arm of the two-factor hypothesis: when the cytosol cannot dehydrate quickly enough, it supercools, and ice nucleates *inside* the cell.

Intracellular supercooling

The intracellular liquidus is set by the cytosol's own osmolality through equation (4); the supercooling is how far the cell temperature has fallen below it:

$$T_{m,i} = -\Delta T_f(\text{osm}_i) \quad ; \quad \Delta T_{sc} = T_{m,i} - T \quad (6)$$

Surface-catalysed nucleation (SCN)

Intracellular ice forms by heterogeneous nucleation catalysed at the inner plasma-membrane surface by the adjacent external ice — the Toner mechanism.^{7,8} The nucleation rate per cell rises catastrophically with supercooling:

$$J = \Omega \cdot A \cdot \phi_w \cdot (\mu)^n \cdot \exp(-\kappa / (T^3 \cdot \Delta T_{sc}^2)) \quad (7)$$

where Ω and κ are the kinetic and thermodynamic SCN parameters, A the membrane area, ϕ_w the freezable-water fraction and μ the molecular mobility (§5). The cumulative probability of at least one nucleation event integrates the rate over time:

$$P_{iif} = 1 - \exp(-\int (J + J_{prop}) dt) \quad (8)$$

The exponential in equation (7) is why IIF has a threshold character: small supercooling gives a negligible rate; once ΔT_{sc} is large the rate explodes and $P_{iif} \rightarrow 1$. A cryoprotectant suppresses this two ways — it keeps the cell dehydrated (small ΔT_{sc}) and it lowers the freezable-water fraction ϕ_w — so with adequate CPA the membrane stays a barrier and P_{iif} stays near zero. The interface visualizes ΔT_{sc} as the gap between the supercooled intracellular point and the liquidus on the phase-diagram inset, and P_{iif} as the **intracellular ice crystals** that nucleate at the membrane (SCN) and grow inward, with breach markers where external ice penetrates.

Propagation and organelle nucleation

Two additional hazards feed equation (8). **Intercellular propagation** spreads ice through cell-cell junctions, autocatalytically — one nucleation event can sweep a confluent sheet,^{9,10} so monolayers suffer far more IIF than dissociated suspensions (junction = 0 for suspensions, preserving the benchmark):

$$J_{prop} = k_{prop} \cdot \text{junction} \cdot \mu \cdot (P_{iif} + 0.004 \cdot f_{ice}) \quad (9)$$

And the **mitochondrion** acts as an independent nucleator (§7). IIF lethality scales with the peak supercooling at nucleation — deeper supercooling yields more, finer, more damaging crystals — saturating to fully lethal when slow warming permits recrystallisation.⁷

05 Membrane phase & vitrification

The bilayer undergoes the **liquid-crystalline** → **gel** ($L_\alpha \rightarrow L_\beta$) transition below its melting temperature T_m . The gel fraction is a sigmoid in temperature, with T_m and the transition width shifted by cholesterol and CPA:

$$\text{gel}(T) = 1 / (1 + \exp((T - T_m) / (w/4))) \quad (10)$$

Above T_m the membrane is a fluid with thermal undulations; below it the bilayer is a two-dimensional solid — undulation amplitude scales as $\sqrt{(k_B T / \kappa_{\text{bend}})}$, bending rigidity jumps, and lateral remodelling ceases.¹¹ In the interface this is enforced literally: as **gel** rises the deformable membrane's thermal noise and velocity are quenched and, above $\text{gel} \approx 0.8$, the contour *locks* — a stored cell holds its crenated shape as a rigid shell and re-fluidizes only on warming back through T_m .

Cytosolic vitrification

Molecular mobility μ follows a WLF/ T_g ' law;⁶ it gates every reaction rate and the nucleation term in equation (7). Its complement is the glass fraction:

$$\text{glass} = 1 - \mu(T; T_g') \quad (11)$$

As the cytosol vitrifies it rigidifies the whole cell and locks the organelles in place — a glass has no long-range diffusion. Storage below T_g' arrests all molecular motion, which is why deep-cold storage is stable.

06 Cytoskeleton & mechanics

The cytoskeleton is modelled as a prestressed **tensegrity** network:¹² microtubules bear compression (struts), the actin cortex and intermediate filaments bear tension (cables). Element-wise stress σ then drives biology rather than a single aggregate tension:

- **Microtubules buckle** under the compressive load of dehydration and the inter-grain ice squeeze, releasing GEF-H1 and driving $\text{RhoA} \rightarrow \text{ROCK} \rightarrow \text{pMLC}$ *before* the network fully depolymerises.
- **Actin and microtubules are cold-labile** — both depolymerise on cooling (MT strongly, F-actin mildly) and cannot repolymerise until the cytosol de-vitrifies on rewarming (repair gated by μ). **Vimentin** intermediate filaments are cold-stable and become the load-bearing safety net against lytic swelling. The interface draws each filament class dissolving into free monomers as its polymerized fraction falls.
- **Mechanical stress reaches the mitochondria**, reversibly depressing $\Delta\Psi_m$ (to ~65% during the mechanically-stressed freeze) and adding a mechanosensitive term to the permeability transition.

This chain is drawn end-to-end in the FA-LINC panel: ECM $\rightarrow$ integrins $\rightarrow$ talin/vinculin/FAK $\rightarrow$ actin stress fibre $\rightarrow$ RhoA-ROCK-LIMK-cofilin $\rightarrow$ nesprin-SUN $\rightarrow$ lamin A/C $\rightarrow$ mechanosensitive genes (YAP/TAZ).

07 Organelles

The **mitochondrion** is an independent osmometer and ice nucleator. Its inner membrane exchanges water fast (aquaporin-8), so the matrix stays near cytosolic osmolality and supercools with it — but the cardiolipin-rich inner membrane is an efficient heterogeneous nucleator, so per unit area it fires equation (7) more readily than the plasma membrane.^{13,14} A frozen mitochondrion seeds cytosolic ice and opens the permeability transition directly, which is what makes fast cooling markedly worse. CPA crosses the inner membrane with a lag; at slow cooling the matrix pre-equilibrates and is protected, at fast cooling it stays under-protected and freezes first.

Organelles are drawn at their true physical size relative to the whole cell (a mitochondrion $\sim 2.1 \times 0.6 \mu\text{m}$ in a $\sim 15 \mu\text{m}$ cell) via the fixed pixel-per-micron scale, and shrink mildly with dehydration. The **nucleus** is deliberately passive: nuclear-pore permeability keeps it in equilibrium with the cytosol, so it tracks the cytosol rather than acting as an independent osmometer — the physically correct choice, not a shortcut.

08 Injury & survival

Immediate survival is the product of five independent survival factors — one per biophysical injury channel — so any single lethal insult caps the outcome:

$$S_{\text{imm}} = (1 - D_{\text{iif}})(1 - D_{\text{osm}})(1 - D_{\text{mem}})(1 - D_{\text{tox}})(1 - D_{\text{mech}})$$

(12)

TERM	INJURY	DRIVER
D _{iif}	intracellular ice	P _{iif} (eq. 8) + mitochondrial ice, scaled by supercooling
D _{osm}	osmotic / solution-effects	freeze-concentrated salt exposure; excursions past volume limits
D _{mem}	membrane lysis	pore/defect fraction; fragmentation
D _{tox}	CPA toxicity	Arrhenius-gated denaturation at high warm CPA concentration
D _{mech}	mechanical	inter-grain channel squeeze; cytoskeletal overload

A regulated-death layer then acts on the survivors. The 24 h endpoint subtracts apoptosis and necrosis; the settled 72 h endpoint subtracts a **delayed wave** — cryopreservation-induced delayed-onset cell death (CIDOC), driven by residual ROS and primed executioner caspase that keep killing cells for one to three days after thaw:^{15,16}

$$S_{24} = S_{\text{imm}} \cdot (1 - D_{\text{apop}})$$

(13)

$$S_{72} = S_{24} \cdot (1 - D_{\text{delayed}}), \quad D_{\text{delayed}} = 0.55 \cdot ROS_{\text{peak}} + 0.35 \cdot \text{casp3} \quad (14)$$

Functional recovery F_{rec} gates surviving cells further by the machinery they need to work again — repolymerized cytoskeleton, restored $\Delta\Psi_m$ and ATP, re-established adhesion, and correctly folded protein. A cell can survive and still fail to function; the model reports both. Cell-line robustness (apoptosis / anoikis / antioxidant / glycolytic axes) acts only on this regulated-death layer — a cell full of ice is dead regardless of genotype, so S_{imm} caps every line alike.

09 Interface mapping

Nothing in the view is decorative. Each element is a projection of a state variable the engine integrates.

YOU SEE	IT IS
Cell size & crenation	V/V_{iso} and membrane slack from KK transport (eq. 1–3)
Water-flux arrows	sign and magnitude of dV_w/dt (eq. 1)
Extracellular CPA/salt density	freeze-concentration factor, Os_{me} (eq. 5)
Phase-diagram inset	operating point & ΔT_{SC} gap on the liquidus (eq. 4, 6)
Intracellular ice crystals	P_{iif} via SCN; nucleate at membrane, grow inward (eq. 7–8)
Rigid vs. wobbling membrane	gel fraction (eq. 10) — locks below T_m
Glassy sheen over the interior	vitrified fraction (eq. 11)
Filaments dissolving to monomers	polymerized fractions mt / $actin$ / $intf$
Timeline: amber + blue curves	temperature and cell volume vs. protocol time
Outcome / cryo-signatures	S_{imm} , S_{24} , S_{72} , F_{rec} and the five D-terms (eq. 12–14)

Render modes (Dark, Illustrative, Fluorescence, Phase-contrast) re-map the palette without changing the physics; the scientific overlay adds leader-line labels, the scale bar, and the phase diagram. Layer toggles switch individual compartments on and off, the way a confocal channel picker does.

10 Calibration & honest limits

The model is anchored to one well-characterised system: **hMSC + DMSO at 1 °C/min → 35.4% 24-hour survival**. Against that single anchor it reproduces, without being fitted to any of them, the inverted-U survival curve with an optimum near 1 °C/min; the SCN cooling-rate crossover ($P_{\text{iff}} \approx 0.08$ at 1 °C/min rising to 1.0 by ≥ 10 °C/min); the biphasic Y-27632 dose response; z-VAD-fmk as the strongest single intervention; the protection of a 4 °C post-thaw hold; and combination results such as 7% DMSO + 5% trehalose $\approx$ 10% DMSO at lower toxicity.

SCOPE

The new mechanisms are correct in *form* and literature-sourced in their *parameters*, but the per-CPA σ , E_a , B , T_g' , and the propagation, SCN and ROS constants are literature-typical values, not measurements of a specific line. Cross-cell-type predictive accuracy is unproven and requires bench data the model cannot supply. Three fitted coefficients have nothing behind them (the mobility-damage exponent, the grain-growth constant, the mechanical-squeeze coefficient); the built-in *Measure next?* tool ranks which parameter to calibrate first.

It is a mechanistic model with real parameters, meant to rank protocols and reason about mechanism. Calibrate before quoting a number.

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